

Auteur: Dara van Engelen
Studierichting: Informatica
Oefeningenreeks 5B nr. 1.9

$$\begin{aligned} I &= \int_1^{+\infty} \frac{dx}{x\sqrt{x}} \\ \int \frac{dx}{x\sqrt{x}} &= \int \frac{dx}{x^{\frac{3}{2}}} = \int x^{\frac{-3}{2}} dx = \frac{x^{\frac{-1}{2}}}{\frac{-1}{2}} + C = \frac{-2}{\sqrt{x}} + C \\ I &= \lim_{b \rightarrow +\infty} \left[\frac{-2}{\sqrt{x}} \right]_1^b = \lim_{b \rightarrow +\infty} \left(\frac{-2}{\sqrt{b}} + 2 \right) = 2 \Rightarrow \text{Convergent} \end{aligned}$$

References